Stefan Busch

Østerbyvej 54, 6470 Sydals, +45 91 99 06 83, stefan.busch.sbs@gmail.com
www.linkedin.com/in/stefan-busch-b3027a261

SUMMARY

Software developer and architect with around 15 years of automated testing experience, including FDA-aligned validation. Combines test framework design, AI-assisted testing and requirements traceability with hands-on C#, C++ and Python experience. Brings practical test management and cross-team mentoring to software verification.

KEY ACHIEVEMENTS

- Designed cross-technology test frameworks and improved regression stability at Roche
- Built automated unit and integration testing frameworks and CI/CD tooling at MAN
- Applied AI-assisted test creation and execution at MAN and in two recent client projects
- Maintained requirements-to-test traceability matrices and mentored developers

SELECTED FREELANCE WORK EXPERIENCE

Overlapping project dates indicate concurrent engagements.

04/2026 – present **Confidential client**

Senior Fullstack Developer

Management software for safety-critical systems: AI-assisted test creation and execution, with requirements-to-test traceability matrices.

- Applied DDD and TDD; derived automated tests from domain requirements and linked test results
- Designed feature-slice Clean Architecture, microservices and a Blazor AI portal

09/2015 – 12/2025 **MAN Truck & Bus,** Munich

Senior Software Developer / Software Architect

Simulation software and modular microservice platform; automated testing and CI/CD.

- Designed unit and integration test frameworks; established Git/Jenkins CI/CD tooling
- Used DDD and traceability matrices; applied AI-assisted testing in the later project phase
- Coached developers in testing, mocking and code reviews; aligned quality standards

01/2013 – 08/2014 **Pixel,** Munich

Senior Software Developer

Solar installation planning software: UI development and test management.

- Created test plans; executed unit, integration and system tests and reported results

05/2011 – 12/2012 **Rohde & Schwarz,** Munich

Senior Software Developer

Software-defined radio framework, C++ code generation and target-device deployment.

- Developed automated unit and integration tests; deployed and tested software on target devices

01/2010 – 06/2010 **T-Systems,** Munich

Software Developer

BizTalk deployment automation across approximately 50 server environments.

- Developed deployment tooling; performed integration tests and post-deployment validation

SELECTED CONSULTING PROJECTS

10/2025 – 03/2026 **Confidential client — Aerospace research, Germany**

AI System Architect, Prompt Engineer, Context Engineer

State-oriented AI agents and multi-agent orchestration; Python, .NET 10 and Ollama.

- Applied AI-assisted test creation and execution, with review and validation of generated test code
- Applied DDD; maintained traceability matrices linking requirements, tests and results

03/2020 – 09/2023 **Roche,** Penzberg

Test Manager, Senior Software Developer

Medical and laboratory software: FDA-aligned test strategies and system validation.

- Designed a cross-technology test framework; automated UI and integration tests with Selenium
- Evaluated test results and improved regression stability with developers and QA
- Assessed verification results and remaining software risks to support release decisions

11/2017 – 02/2019 **ASM Assembly Systems,** Munich

Senior Software Developer, QA Manager

Material logistics services for PCB assembly robots; cross-system quality assurance.

- Defined test strategy; implemented unit and integration tests and reported test results
- Developed RPC services and executable cross-system runbooks

MORE PROJECTS **www.linkedin.com/in/stefan-busch-b3027a261**

SKILLS

Verification & Validation

- Verification Planning & Test Cases
- Requirements Traceability Matrices
- Unit, Integration & System Testing
- Regression & UI Test Automation
- FDA-Aligned Software Validation
- Risk Assessment & Test Reporting

Engineering & Collaboration

- Technical Mentoring & Code Reviews
- Cross-Team Quality Standards
- Agile / Scrum
- Domain-Driven Design (DDD)
- Clean Code, SOLID & TDD

Programming & Test Tools

- C#, .NET / .NET Framework
- C / C++, Python
- NUnit, MSTest, Selenium
- Postman / Newman

AI-Assisted Testing

- Test Creation & Execution
- Generated Test Code Validation
- Prompt & Context Engineering
- AI Agents & Orchestration

Platforms & Integration

- Microservices, REST, Blazor
- PostgreSQL, Oracle, SQL Server
- Docker, Linux & Windows

CI/CD & Tooling

- Git, Jenkins, TeamCity
- Deployment Automation
- Test Environment Configuration

EDUCATION

10/1985 – 08/1990 **Diploma** in Technical Physics, FH Munich

10/1981 – 05/1984 **Journeyman's certificate**, Central Heating & Ventilation Engineering

LANGUAGES

German	English	Danish
C2	C1	A1 (ongoing studies at UCplus Danish Language School)